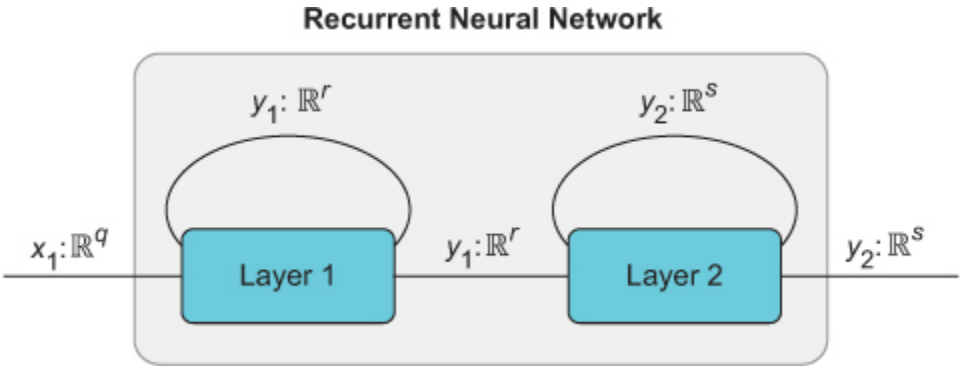


Now we can see the inside the original process box, and it is composed of its own set of subprocesses. We can see that our *n*-dimensional vector gets multiplied by a matrix of dimensions $n \times m$, which produces an *m*-dimensional vector product. This vector then passes through some process called “ReLU,” which you may recognize as a standard neural network activation function, the rectified linear unit. We could continue to zoom in on the ReLU sub-subprocess if we wanted. Anything that deserves the name *string diagram* must be able to be scrutinized at any level of abstraction and remain *well typed* at any level (meaning the types of the data entering and exiting the processes must be compatible and make sense—a process that is supposed to produce sorted lists should not be hooked up to another process that expects integers).

As long as the strings are well typed, we can string together a bunch of processes into a complex system. This allows us to build components once and re-use them wherever they’re type-matched. At a somewhat high level, we might depict a simple two-layer recurrent neural network (RNN) like this:



This RNN takes in a *q* vector and produces an *s* vector. However, we can see the inside processes. There are two layers, and each one looks identical in its function. They each take in a vector and produce a vector, except that